

Not a Typical Firm: The Joint Dynamics of Firms, Labor Shares, and Capital-Labor Substitution

Hubmer & Restrepo (2021, WP)

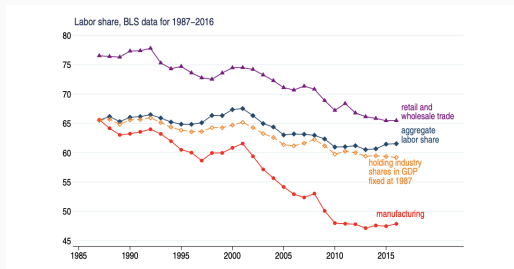
Felipe Camêlo

October 13th 2021, New York University, BGGM Reading Group

Introduction - The Decline in the US Labor Share

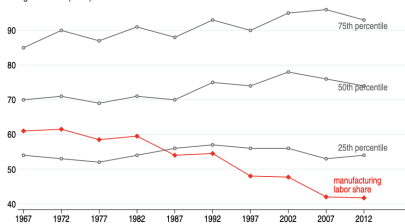
Two broad explanations:

- **Technology:** substitution of capital for labor in widening range of tasks as automation advances and capital becomes cheaper.
(Karabarbounis– Neiman 2014; Edden–Gagl 2018; Hubmer 2020; Acemoglu–Restrepo, 2018)
- **Concentration:** rising competition reallocates economic activity to high markup firms
(Barkai 2020; De Loecker–Eeckhout–Unger 2020; Autor–Dorn–Katz–Patterson–V.Reenen 2020)



Introduction - The Role of Firms in the Decline of the Labor Share

Labor share decomposition in manufacturing
Kehrig & Vincent (2020)



- Decline not uniform across firms, and not visible for median (*typical*) firm (Autor et al. 2020; Kehrig–Vincent, 2020)
- Challenge to the technology view Concentration:
 - rules out simple story where all firms face same factor prices and have access to same technologies
 - suggests key role for reallocation rather than capital-labor substitution

This paper

Underlying Question → What is the right framework to make sense of these facts jointly?

1. Develop firm dynamics model with **costly capital–labor substitution within tasks**: matches firm-level labor and market share dynamics as capital prices falling
 - typical firm: no change in task allocation $\implies K$ and L complements $\implies$ LS rises
 - top firms: more tasks automated $\implies K$ and L substitutes $\implies$ LS declines
2. Extend to non-CES demand system to account for **rising competition** and reallocation to more productive firms: K–L substitution remains important
 - 90% of the decline in labor share in manufacturing
 - 40% of the decline in labor share in retail; rising competition more important
3. “Model-free” accounting of contribution of markups/reallocation to LS decline
 - important to allow for **differences in technology across firms** (cf. DeLoecker, Eeckhout, Unger 2020)

1. **The Decline in the Labor Share: Measurement and explanations**

Karabarbounis and Neiman (2014), Elsby, Hobijn and Sahin, (2013), Rognlie (2015), Bridgman (2018), Smith, Yagan, Zidar and Zwick (2019) and Koh, Santaaulàlia-Llopis and Zheng (2020)

- Firm-level data to understand this phenomenon across the firm distribution: Autor et al. (2020), Kehrig and Vincent (2020)

2. **The Cause and Effects of Automation**

Acemoglu and Restrepo (2018, 2020), Acemoglu, Lelarge and Restrepo (2020), Eden and Gaggl (2018) and Hubmer (2020)

3. **The Cause and Effects of Concentration**

Barkai (2020), De Loecker, Eeckhout and Unger (2020), Baqaee and Farhi (2020)

- Variations in markup structure: CES vs non-CES Demand Systems – Edmond, Midrigan and Xu (2021)

Model with CES demand to show implications of capital–labor substitution across firms

Model with non-CES demand to compare effects of competition and capital–labor substitution

Model-free bounds on effect of reallocation and changes in markups on labor share

**Model with CES demand to show
implications of capital–labor
substitution across firms**

Model – Overview

Standard firm dynamics model with task level substitution and costly automation

- Firms differ in productivity z (exogenous) and capital share parameter α (endogenous)
 $\implies$ Firms can be summarized by their types $\theta = (z, \alpha)$
- To raise α , firms have to pay a (constant) fixed cost for each automated task
- Matches studies on adoption of new capital-intensive techs (Lashkari et al. 2019; Acemoglu et al. 2020; US Census 2020)
- New technologies diffuse in the economy through imitation by entrants

Model – Production Function and Demand

- A type θ firm produces output using a continuum of tasks with task substitution elasticity $\eta \geq 0$

$$y(\theta) = z \cdot \left(\int_0^1 \mathcal{Y}(\theta, x)^{\frac{\eta-1}{\eta}} dx \right)^{\frac{\eta}{\eta-1}}$$

- Tasks in $[0, \alpha]$ are automated and can be produced by capital

$$\mathcal{Y}(\theta, x) = \begin{cases} \psi_k(x) k(\theta, x) + \psi_l(x) l(\theta, x) & , \text{if } x \leq \alpha \\ \psi_l(x) l(\theta, x) & , \text{if } x > \alpha \end{cases} \quad \text{where } \frac{\psi_l(x)}{q(x) \cdot \psi_k(x)} \text{ increasing in } x$$

so that the firm maximizes $\pi(\theta) = y(\theta) - w \int_0^1 l(\theta, x) dx - \int_0^1 [1/q(x)] k(\theta, x) dx$
choosing $\{l(\theta, x), k(\theta, x)\}_{x \in [0, 1]}$

- Demand is given by a CES system so that

$$y = \left(\int_{\theta} y(\theta)^{\frac{\sigma-1}{\sigma}} m(\theta) d\theta \right)^{\frac{\sigma}{\sigma-1}}$$

Model – Production Function and Demand

- A type θ firm produces output using a continuum of tasks with task substitution elasticity $\eta \geq 0$

$$y(\theta) = z \cdot \left(\int_0^1 \mathcal{Y}(\theta, x)^{\frac{\eta-1}{\eta}} dx \right)^{\frac{\eta}{\eta-1}}$$

- Tasks in $[0, \alpha]$ are automated and can be produced by capital

$$\mathcal{Y}(\theta, x) = \begin{cases} \psi_k(x) k(\theta, x) + \psi_l(x) l(\theta, x) & , \text{if } x \leq \alpha \\ \psi_l(x) l(\theta, x) & , \text{if } x > \alpha \end{cases} \quad \text{where } \frac{\psi_l(x)}{q(x) \cdot \psi_k(x)} \text{ increasing in } x$$

so that the firm maximizes $\pi(\theta) = y(\theta) - w \int_0^1 l(\theta, x) dx - \int_0^1 [1/q(x)] k(\theta, x) dx$
choosing $\{l(\theta, x), k(\theta, x)\}_{x \in [0, 1]}$

- Demand is given by a CES system so that

$$y = \left(\int_{\theta} y(\theta)^{\frac{\sigma-1}{\sigma}} m(\theta) d\theta \right)^{\frac{\sigma}{\sigma-1}}$$

Model – Production Function and Demand

- A type θ firm produces output using a continuum of tasks with task substitution elasticity $\eta \geq 0$

$$y(\theta) = z \cdot \left(\int_0^1 \mathcal{Y}(\theta, x)^{\frac{\eta-1}{\eta}} dx \right)^{\frac{\eta}{\eta-1}}$$

- Tasks in $[0, \alpha]$ are automated and can be produced by capital

$$\mathcal{Y}(\theta, x) = \begin{cases} \psi_k(x) k(\theta, x) + \psi_l(x) l(\theta, x) & , \text{if } x \leq \alpha \\ \psi_l(x) l(\theta, x) & , \text{if } x > \alpha \end{cases} \quad \text{where } \frac{\psi_l(x)}{q(x) \cdot \psi_k(x)} \text{ increasing in } x$$

so that the firm maximizes $\pi(\theta) = y(\theta) - w \int_0^1 l(\theta, x) dx - \int_0^1 [1/q(x)] k(\theta, x) dx$
choosing $\{l(\theta, x), k(\theta, x)\}_{x \in [0, 1]}$

- Demand is given by a CES system so that

$$y = \left(\int_{\theta} y(\theta)^{\frac{\sigma-1}{\sigma}} m(\theta) d\theta \right)^{\frac{\sigma}{\sigma-1}}$$

Model – Timing for Incumbents

- Incumbent firms begin a period with productivity z and automation level α
- They then make optimal employment and capital utilization decisions and collect profits $\pi(\theta)$
- Subsequently, firms draw a fixed operating cost $c_f \cdot y$, where $c_f \sim G(c_f)$ and decide whether to continue operating
- If they continue, they can automate additional tasks in $(\alpha, \alpha']$ at a cost $c_a \cdot y (\alpha' - \alpha)$ and draw next period's productivity z' , which follows an exogenous AR(1) in logs.

Model – Timing for Incumbents

- Incumbent firms begin a period with productivity z and automation level α
- They then make optimal employment and capital utilization decisions and collect profits $\pi(\theta)$
- Subsequently, firms draw a fixed operating cost $c_f \cdot y$, where $c_f \sim G(c_f)$ and decide whether to continue operating
- If they continue, they can automate additional tasks in $(\alpha, \alpha']$ at a cost $c_a \cdot y (\alpha' - \alpha)$ and draw next period's productivity z' , which follows an exogenous AR(1) in logs.

Model – Timing for Incumbents

- Incumbent firms begin a period with productivity z and automation level α
- They then make optimal employment and capital utilization decisions and collect profits $\pi(\theta)$
- Subsequently, firms draw a fixed operating cost $c_f \cdot y$, where $c_f \sim G(c_f)$ and decide whether to continue operating
- If they continue, they can automate additional tasks in $(\alpha, \alpha']$ at a cost $c_a \cdot y (\alpha' - \alpha)$ and draw next period's productivity z' , which follows an exogenous AR(1) in logs.

Model – Timing for Incumbents

- Incumbent firms begin a period with productivity z and automation level α
- They then make optimal employment and capital utilization decisions and collect profits $\pi(\theta)$
- Subsequently, firms draw a fixed operating cost $c_f \cdot y$, where $c_f \sim G(c_f)$ and decide whether to continue operating
- If they continue, they can automate additional tasks in $(\alpha, \alpha']$ at a cost $c_a \cdot y (\alpha' - \alpha)$ and draw next period's productivity z' , which follows an exogenous AR(1) in logs.

Model – Timing for Entrants

- Every period a unit mass of potential entrants decides whether to enter the market
- Entrants draw a productivity signal z from a distribution $H(z)$ and start with a common level of automation α_0
- After observing z , 0 , as well as the realization of the fixed operating cost c_f , entrants decide whether to pay the fixed cost and enter
- After paying this cost, entrants become incumbents and face the same problem described in the last slide
- Finally, it is assumed that automation technologies also have some exogenous diffusion through entrants, and that α_0 equals the unweighted average of α among incumbents

Model – Timing for Entrants

- Every period a unit mass of potential entrants decides whether to enter the market
- Entrants draw a productivity signal z from a distribution $H(z)$ and start with a common level of automation α_0
- After observing z , 0 , as well as the realization of the fixed operating cost c_f , entrants decide whether to pay the fixed cost and enter
- After paying this cost, entrants become incumbents and face the same problem described in the last slide
- Finally, it is assumed that automation technologies also have some exogenous diffusion through entrants, and that α_0 equals the unweighted average of α among incumbents

Model – Timing for Entrants

- Every period a unit mass of potential entrants decides whether to enter the market
- Entrants draw a productivity signal z from a distribution $H(z)$ and start with a common level of automation α_0
- After observing z , 0 , as well as the realization of the fixed operating cost c_f , entrants decide whether to pay the fixed cost and enter
- After paying this cost, entrants become incumbents and face the same problem described in the last slide
- Finally, it is assumed that automation technologies also have some exogenous diffusion through entrants, and that α_0 equals the unweighted average of α among incumbents

Model – Timing for Entrants

- Every period a unit mass of potential entrants decides whether to enter the market
- Entrants draw a productivity signal z from a distribution $H(z)$ and start with a common level of automation α_0
- After observing z , θ , as well as the realization of the fixed operating cost c_f , entrants decide whether to pay the fixed cost and enter
- After paying this cost, entrants become incumbents and face the same problem described in the last slide
- Finally, it is assumed that automation technologies also have some exogenous diffusion through entrants, and that α_0 equals the unweighted average of α among incumbents

Model – Timing for Entrants

- Every period a unit mass of potential entrants decides whether to enter the market
- Entrants draw a productivity signal z from a distribution $H(z)$ and start with a common level of automation α_0
- After observing z , θ , as well as the realization of the fixed operating cost c_f , entrants decide whether to pay the fixed cost and enter
- After paying this cost, entrants become incumbents and face the same problem described in the last slide
- Finally, it is assumed that automation technologies also have some exogenous diffusion through entrants, and that α_0 equals the unweighted average of α among incumbents

Model – Value Functions for Entrants and Incumbents

- Both incumbents and new entrants discount the future at a constant interest rate r , which is taken as exogenous

- Value function of incumbent with type θ

$$V(\theta) = \pi(\theta) + \int \max \left\{ 0, -c_f \cdot y + \max_{\alpha' \in [\alpha, 1]} \left\{ -c_a \cdot (\alpha' - \alpha) + \frac{1}{1+r} \mathbb{E} \left[V(\theta') \mid z, \alpha' \right] \right\} \right\} dG(c_f)$$

- Value function of (potential) entrants with productivity signal z

$$V^e(\theta) = \int \max \left\{ 0, -c_f \cdot y + \max_{\alpha' \in [\alpha_0, 1]} \left\{ -c_a \cdot (\alpha' - \alpha_0) + \frac{1}{1+r} \mathbb{E} \left[V(\theta') \mid z, \alpha' \right] \right\} \right\} dG(c_f)$$

- Diffusion of automation through imitation: $a_0 = \bar{\alpha}$

- Exogenous process for productivity: $\log z' = \rho_z \log z + \varepsilon', \varepsilon' \sim N(\mu_z, \sigma_z)$

Model – Value Functions for Entrants and Incumbents

- Both incumbents and new entrants discount the future at a constant interest rate r , which is taken as exogenous

- Value function of incumbent with type θ

$$V(\theta) = \pi(\theta) + \int \max \left\{ 0, -c_f \cdot y + \max_{\alpha' \in [\alpha, 1]} \left\{ -c_a \cdot (\alpha' - \alpha) + \frac{1}{1+r} \mathbb{E} \left[V(\theta') \mid z, \alpha' \right] \right\} \right\} dG(c_f)$$

- Value function of (potential) entrants with productivity signal z

$$V^e(\theta) = \int \max \left\{ 0, -c_f \cdot y + \max_{\alpha' \in [\alpha_0, 1]} \left\{ -c_a \cdot (\alpha' - \alpha_0) + \frac{1}{1+r} \mathbb{E} \left[V(\theta') \mid z, \alpha' \right] \right\} \right\} dG(c_f)$$

- Diffusion of automation through imitation: $a_0 = \bar{\alpha}$

- Exogenous process for productivity: $\log z' = \rho_z \log z + \varepsilon', \varepsilon' \sim N(\mu_z, \sigma_z)$

Model – Value Functions for Entrants and Incumbents

- Both incumbents and new entrants discount the future at a constant interest rate r , which is taken as exogenous

- Value function of incumbent with type θ

$$V(\theta) = \pi(\theta) + \int \max \left\{ 0, -c_f \cdot y + \max_{\alpha' \in [\alpha, 1]} \left\{ -c_a \cdot (\alpha' - \alpha) + \frac{1}{1+r} \mathbb{E} \left[V(\theta') \mid z, \alpha' \right] \right\} \right\} dG(c_f)$$

- Value function of (potential) entrants with productivity signal z

$$V^e(\theta) = \int \max \left\{ 0, -c_f \cdot y + \max_{\alpha' \in [\alpha_0, 1]} \left\{ -c_a \cdot (\alpha' - \alpha_0) + \frac{1}{1+r} \mathbb{E} \left[V(\theta') \mid z, \alpha' \right] \right\} \right\} dG(c_f)$$

- Diffusion of automation through imitation: $a_0 = \bar{\alpha}$

- Exogenous process for productivity: $\log z' = \rho_z \log z + \varepsilon', \varepsilon' \sim N(\mu_z, \sigma_z)$

Model – Value Functions for Entrants and Incumbents

- Both incumbents and new entrants discount the future at a constant interest rate r , which is taken as exogenous

- Value function of incumbent with type θ

$$V(\theta) = \pi(\theta) + \int \max \left\{ 0, -c_f \cdot y + \max_{\alpha' \in [\alpha, 1]} \left\{ -c_a \cdot (\alpha' - \alpha) + \frac{1}{1+r} \mathbb{E} \left[V(\theta') \mid z, \alpha' \right] \right\} \right\} dG(c_f)$$

- Value function of (potential) entrants with productivity signal z

$$V^e(\theta) = \int \max \left\{ 0, -c_f \cdot y + \max_{\alpha' \in [\alpha_0, 1]} \left\{ -c_a \cdot (\alpha' - \alpha_0) + \frac{1}{1+r} \mathbb{E} \left[V(\theta') \mid z, \alpha' \right] \right\} \right\} dG(c_f)$$

- Diffusion of automation through imitation: $a_0 = \bar{\alpha}$

- Exogenous process for productivity: $\log z' = \rho_z \log z + \varepsilon', \varepsilon' \sim N(\mu_z, \sigma_z)$

Model – Value Functions for Entrants and Incumbents

- Both incumbents and new entrants discount the future at a constant interest rate r , which is taken as exogenous

- Value function of incumbent with type θ

$$V(\theta) = \pi(\theta) + \int \max \left\{ 0, -c_f \cdot y + \max_{\alpha' \in [\alpha, 1]} \left\{ -c_a \cdot (\alpha' - \alpha) + \frac{1}{1+r} \mathbb{E} \left[V(\theta') \mid z, \alpha' \right] \right\} \right\} dG(c_f)$$

- Value function of (potential) entrants with productivity signal z

$$V^e(\theta) = \int \max \left\{ 0, -c_f \cdot y + \max_{\alpha' \in [\alpha_0, 1]} \left\{ -c_a \cdot (\alpha' - \alpha_0) + \frac{1}{1+r} \mathbb{E} \left[V(\theta') \mid z, \alpha' \right] \right\} \right\} dG(c_f)$$

- Diffusion of automation through imitation: $a_0 = \bar{\alpha}$

- Exogenous process for productivity: $\log z' = \rho_z \log z + \varepsilon', \varepsilon' \sim N(\mu_z, \sigma_z)$

- The authors restrict attention to perfect foresight equilibria
- Given a path for investment productivities $q_t(x)$, an initial distribution of firms $m_0(\theta)$, an equilibrium path for wages w_t and aggregate output y_t , as well as a path for the distribution of firms $m_t(\theta)$ such that for all t :
 1. the ideal-price index condition holds
 2. the labor market clears
 3. automation and exit decisions maximize the value function of incumbents
 4. entry decisions maximize the value of entrants
 5. starting from $m_0(\theta)$, the evolution of $m_t(\theta)$ is governed by the exogenous process for z , the endogenous process for α , and optimal entry and exit decisions of firms.

Model – Equilibrium

- The authors restrict attention to perfect foresight equilibria
- Given a path for investment productivities $q_t(x)$, an initial distribution of firms $m_0(\theta)$, an equilibrium path for wages w_t and aggregate output y_t , as well as a path for the distribution of firms $m_t(\theta)$ such that for all t :
 1. the ideal-price index condition holds
 2. the labor market clears
 3. automation and exit decisions maximize the value function of incumbents
 4. entry decisions maximize the value of entrants
 5. starting from $m_0(\theta)$, the evolution of $m_t(\theta)$ is governed by the exogenous process for z , the endogenous process for α , and optimal entry and exit decisions of firms.

Model – Equilibrium

- The authors restrict attention to perfect foresight equilibria
- Given a path for investment productivities $q_t(x)$, an initial distribution of firms $m_0(\theta)$, an equilibrium path for wages w_t and aggregate output y_t , as well as a path for the distribution of firms $m_t(\theta)$ such that for all t :
 1. the ideal-price index condition holds
 2. the labor market clears
 3. automation and exit decisions maximize the value function of incumbents
 4. entry decisions maximize the value of entrants
 5. starting from $m_0(\theta)$, the evolution of $m_t(\theta)$ is governed by the exogenous process for z , the endogenous process for α , and optimal entry and exit decisions of firms.

Model – Equilibrium

- The authors restrict attention to perfect foresight equilibria
- Given a path for investment productivities $q_t(x)$, an initial distribution of firms $m_0(\theta)$, an equilibrium path for wages w_t and aggregate output y_t , as well as a path for the distribution of firms $m_t(\theta)$ such that for all t :
 1. the ideal-price index condition holds
 2. the labor market clears
 3. automation and exit decisions maximize the value function of incumbents
 4. entry decisions maximize the value of entrants
 5. starting from $m_0(\theta)$, the evolution of $m_t(\theta)$ is governed by the exogenous process for z , the endogenous process for α , and optimal entry and exit decisions of firms.

Model – Equilibrium

- The authors restrict attention to perfect foresight equilibria
- Given a path for investment productivities $q_t(x)$, an initial distribution of firms $m_0(\theta)$, an equilibrium path for wages w_t and aggregate output y_t , as well as a path for the distribution of firms $m_t(\theta)$ such that for all t :
 1. the ideal-price index condition holds
 2. the labor market clears
 3. automation and exit decisions maximize the value function of incumbents
 4. entry decisions maximize the value of entrants
 5. starting from $m_0(\theta)$, the evolution of $m_t(\theta)$ is governed by the exogenous process for z , the endogenous process for α , and optimal entry and exit decisions of firms.

Model – Equilibrium

- The authors restrict attention to perfect foresight equilibria
- Given a path for investment productivities $q_t(x)$, an initial distribution of firms $m_0(\theta)$, an equilibrium path for wages w_t and aggregate output y_t , as well as a path for the distribution of firms $m_t(\theta)$ such that for all t :
 1. the ideal-price index condition holds
 2. the labor market clears
 3. automation and exit decisions maximize the value function of incumbents
 4. entry decisions maximize the value of entrants
 5. starting from $m_0(\theta)$, the evolution of $m_t(\theta)$ is governed by the exogenous process for z , the endogenous process for α , and optimal entry and exit decisions of firms.

Model – Equilibrium

- The authors restrict attention to perfect foresight equilibria
- Given a path for investment productivities $q_t(x)$, an initial distribution of firms $m_0(\theta)$, an equilibrium path for wages w_t and aggregate output y_t , as well as a path for the distribution of firms $m_t(\theta)$ such that for all t :
 1. the ideal-price index condition holds
 2. the labor market clears
 3. automation and exit decisions maximize the value function of incumbents
 4. entry decisions maximize the value of entrants
 5. starting from $m_0(\theta)$, the evolution of $m_t(\theta)$ is governed by the exogenous process for z , the endogenous process for α , and optimal entry and exit decisions of firms.

Model – Equilibrium – Optimal Automation

- Let α_t^* denote the α that minimizes production costs if this could happen at no costs

$$\frac{\psi_l(\alpha_t^*)}{q(\alpha_t^*) \cdot \psi_k(\alpha_t^*)} = w_t$$

- Optimal automation choice is an increasing function of z : $\alpha_{t+1} = \max\{\alpha_t, \hat{\alpha}_t(z)\}$ as
 - $\hat{\alpha}_t(z)$ is an increasing function of z with $\lim_{z \rightarrow 0} \hat{\alpha}_t(z) = 0$ and $\lim_{z \rightarrow \infty} \hat{\alpha}_t(z) = \alpha_{t+1}^*$
- Automation episodes happen as $\hat{\alpha}_t(z_t)$ crosses current α_t for growing and large firms

Model – Equilibrium – Optimal Automation

- Let α_t^* denote the α that minimizes production costs if this could happen at no costs

$$\frac{\psi_l(\alpha_t^*)}{q(\alpha_t^*) \cdot \psi_k(\alpha_t^*)} = w_t$$

- Optimal automation choice is an increasing function of z : $\alpha_{t+1} = \max\{\alpha_t, \hat{\alpha}_t(z)\}$ as
 - $\hat{\alpha}_t(z)$ is an increasing function of z with $\lim_{z \rightarrow 0} \hat{\alpha}_t(z) = 0$ and $\lim_{z \rightarrow \infty} \hat{\alpha}_t(z) = \alpha_{t+1}^*$
- Automation episodes happen as $\hat{\alpha}_t(z_t)$ crosses current α_t for growing and large firms

Model – Equilibrium – Optimal Automation

- Let α_t^* denote the α that minimizes production costs if this could happen at no costs

$$\frac{\psi_l(\alpha_t^*)}{q(\alpha_t^*) \cdot \psi_k(\alpha_t^*)} = w_t$$

- Optimal automation choice is an increasing function of z : $\alpha_{t+1} = \max\{\alpha_t, \hat{\alpha}_t(z)\}$ as
 - $\hat{\alpha}_t(z)$ is an increasing function of z with $\lim_{z \rightarrow 0} \hat{\alpha}_t(z) = 0$ and $\lim_{z \rightarrow \infty} \hat{\alpha}_t(z) = \alpha_{t+1}^*$
- Automation episodes happen as $\hat{\alpha}_t(z_t)$ crosses current α_t for growing and large firms

Model – Equilibrium – Two Key Elasticities

- The capital share of cost and the labor share of costs, which equal the elasticities of output wrt both factors respectively, are given by

$$\varepsilon_k(\alpha, w) = \frac{\Psi_k(\alpha)}{\Psi_k(\alpha) + \Psi_l(\alpha) w^{1-\eta}} \varepsilon_l(\alpha, w) = \frac{\Psi_l(\alpha) w^{1-\eta}}{\Psi_k(\alpha) + \Psi_l(\alpha) w^{1-\eta}}$$

$$\text{where } \Psi_k(\alpha) = \int_0^\alpha (q(x) \psi_k(x))^{\eta-1} dx \text{ and } \Psi_l(\alpha) = \int_\alpha^1 (\psi_l(x))^{\eta-1} dx$$

- Short-run K–L elasticity (fixed α) equals the task substitution elasticity η
 $\implies$ maps to firm-level short-run K–L elasticity

- Long-run K–L elasticity (allows for adjusting α to α^*) equals

$$\eta_{LR} = \eta + \frac{\partial \ln(\Psi_k(\alpha)/\Psi_l(\alpha))}{\partial \alpha} \bigg/ \frac{\partial \ln(\psi_k(\alpha)/\psi_l(\alpha))}{\partial \alpha}$$

- As firms grow and $\hat{\alpha}_{t+1}(z) \rightarrow \alpha_{t+1}^*$, elasticity goes from η to η_{LR}

Model – Equilibrium – Two Key Elasticities

- The capital share of cost and the labor share of costs, which equal the elasticities of output wrt both factors respectively, are given by

$$\varepsilon_k(\alpha, w) = \frac{\Psi_k(\alpha)}{\Psi_k(\alpha) + \Psi_l(\alpha) w^{1-\eta}} \varepsilon_l(\alpha, w) = \frac{\Psi_l(\alpha) w^{1-\eta}}{\Psi_k(\alpha) + \Psi_l(\alpha) w^{1-\eta}}$$

$$\text{where } \Psi_k(\alpha) = \int_0^\alpha (q(x) \psi_k(x))^{\eta-1} dx \text{ and } \Psi_l(\alpha) = \int_\alpha^1 (\psi_l(x))^{\eta-1} dx$$

- **Short-run K–L elasticity** (fixed α) equals the task substitution elasticity η
 $\implies$ maps to firm-level short-run K–L elasticity

- **Long-run K–L elasticity** (allows for adjusting α to α^*) equals

$$\eta_{LR} = \eta + \frac{\partial \ln(\Psi_k(\alpha)/\Psi_l(\alpha))}{\partial \alpha} \bigg/ \frac{\partial \ln(\psi_k(\alpha)/\psi_l(\alpha))}{\partial \alpha}$$

- As firms grow and $\hat{\alpha}_{t+1}(z) \rightarrow \alpha_{t+1}^*$, elasticity goes from η to η_{LR}

Model – Equilibrium – Two Key Elasticities

- The capital share of cost and the labor share of costs, which equal the elasticities of output wrt both factors respectively, are given by

$$\varepsilon_k(\alpha, w) = \frac{\Psi_k(\alpha)}{\Psi_k(\alpha) + \Psi_l(\alpha) w^{1-\eta}} \varepsilon_l(\alpha, w) = \frac{\Psi_l(\alpha) w^{1-\eta}}{\Psi_k(\alpha) + \Psi_l(\alpha) w^{1-\eta}}$$

$$\text{where } \Psi_k(\alpha) = \int_0^\alpha (q(x) \psi_k(x))^{\eta-1} dx \text{ and } \Psi_l(\alpha) = \int_\alpha^1 (\psi_l(x))^{\eta-1} dx$$

- **Short-run K–L elasticity** (fixed α) equals the task substitution elasticity η
 $\implies$ maps to firm-level short-run K–L elasticity

- **Long-run K–L elasticity** (allows for adjusting α to α^*) equals

$$\eta_{LR} = \eta + \frac{\partial \ln(\Psi_k(\alpha)/\Psi_l(\alpha))}{\partial \alpha} \bigg/ \frac{\partial \ln(\psi_k(\alpha)/\psi_l(\alpha))}{\partial \alpha}$$

- As firms grow and $\hat{\alpha}_{t+1}(z) \rightarrow \alpha_{t+1}^*$, elasticity goes from η to η_{LR}

- Standard firm dynamics parametrization to match entry, exit, firm size, markup
- Set technology parameters to match key facts on manufacturing labor share:
 - task-substitution elasticity $\eta = 0.4$: match short-run K-L elasticity (Oberfield-Raval 2020)
 - specify $\psi_l(x) = \left(x^{\frac{1-\eta-\gamma}{\gamma}} - 1\right)^{\frac{1}{1-\eta-\gamma}}$ and $\psi_k(x) = 1$
 $\eta + \gamma$ gives long-run aggregate K - L elasticity: target 1.35 (Hubmer, 2020; Karabarbounis & Neiman, 2014)
 - $d \ln q = 1.39$ to match observed decline in aggregate mf. labor share 1982–2012 (−17.8pp)
 - choose automation cost c_a to target change in median labor share (+3pp)
- What happens if we impose a uniform decrease in the rental price of capital (increase in the return of investment opp. $q(x)$) ?

- Standard firm dynamics parametrization to match entry, exit, firm size, markup
- Set technology parameters to match key facts on manufacturing labor share:
 - task-substitution elasticity $\eta = 0.4$: match short-run K-L elasticity (Oberfield-Raval 2020)
 - specify $\psi_l(x) = \left(x^{\frac{1-\eta-\gamma}{\gamma}} - 1\right)^{\frac{1}{1-\eta-\gamma}}$ and $\psi_k(x) = 1$
 $\eta + \gamma$ gives long-run aggregate K – L elasticity: target 1.35 (Hubmer, 2020; Karabarbounis & Neiman, 2014)
 - $d \ln q = 1.39$ to match observed decline in aggregate mf. labor share 1982–2012 (–17.8pp)
 - choose automation cost c_a to target change in median labor share (+3pp)
- What happens if we impose a uniform decrease in the rental price of capital (increase in the return of investment opp. $q(x)$) ?

- Standard firm dynamics parametrization to match entry, exit, firm size, markup
- Set technology parameters to match key facts on manufacturing labor share:
 - task-substitution elasticity $\eta = 0.4$: match short-run K-L elasticity (Oberfield-Raval 2020)
 - specify $\psi_l(x) = \left(x^{\frac{1-\eta-\gamma}{\gamma}} - 1\right)^{\frac{1}{1-\eta-\gamma}}$ and $\psi_k(x) = 1$
 $\eta + \gamma$ gives long-run aggregate K – L elasticity: target 1.35 (Hubmer, 2020; Karabarbounis & Neiman, 2014)
 - $d \ln q = 1.39$ to match observed decline in aggregate mf. labor share 1982–2012 (–17.8pp)
 - choose automation cost c_a to target change in median labor share (+3pp)
- What happens if we impose a uniform decrease in the rental price of capital (increase in the return of investment opp. $q(x)$) ?

- Standard firm dynamics parametrization to match entry, exit, firm size, markup
- Set technology parameters to match key facts on manufacturing labor share:
 - task-substitution elasticity $\eta = 0.4$: match short-run K-L elasticity (Oberfield-Raval 2020)
 - specify $\psi_l(x) = \left(x^{\frac{1-\eta-\gamma}{\gamma}} - 1\right)^{\frac{1}{1-\eta-\gamma}}$ and $\psi_k(x) = 1$
 $\eta + \gamma$ gives long-run aggregate K – L elasticity: target 1.35 (Hubmer, 2020; Karabarbounis & Neiman, 2014)
 - $d \ln q = 1.39$ to match observed decline in aggregate mf. labor share 1982–2012 (–17.8pp)
 - choose automation cost c_a to target change in median labor share (+3pp)
- What happens if we impose a uniform decrease in the rental price of capital (increase in the return of investment opp. $q(x)$) ?

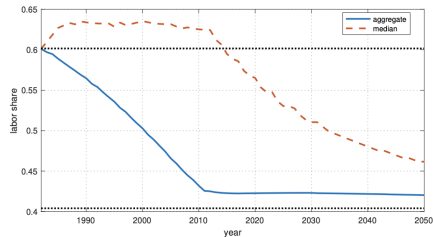
- Standard firm dynamics parametrization to match entry, exit, firm size, markup
- Set technology parameters to match key facts on manufacturing labor share:
 - task-substitution elasticity $\eta = 0.4$: match short-run K-L elasticity (Oberfield-Raval 2020)
 - specify $\psi_l(x) = \left(x^{\frac{1-\eta-\gamma}{\gamma}} - 1\right)^{\frac{1}{1-\eta-\gamma}}$ and $\psi_k(x) = 1$
 $\eta + \gamma$ gives long-run aggregate K – L elasticity: target 1.35 (Hubmer, 2020; Karabarbounis & Neiman, 2014)
 - $d \ln q = 1.39$ to match observed decline in aggregate mf. labor share 1982–2012 (–17.8pp)
 - choose automation cost c_a to target change in median labor share (+3pp)
- What happens if we impose a uniform decrease in the rental price of capital (increase in the return of investment opp. $q(x)$) ?

- Standard firm dynamics parametrization to match entry, exit, firm size, markup
- Set technology parameters to match key facts on manufacturing labor share:
 - task-substitution elasticity $\eta = 0.4$: match short-run K-L elasticity (Oberfield-Raval 2020)
 - specify $\psi_l(x) = \left(x^{\frac{1-\eta-\gamma}{\gamma}} - 1 \right)^{\frac{1}{1-\eta-\gamma}}$ and $\psi_k(x) = 1$
 $\eta + \gamma$ gives long-run aggregate K – L elasticity: target 1.35 (Hubmer, 2020; Karabarbounis & Neiman, 2014)
 - $d \ln q = 1.39$ to match observed decline in aggregate mf. labor share 1982–2012 (–17.8pp)
 - choose automation cost c_a to target change in median labor share (+3pp)
- What happens if we impose a uniform decrease in the rental price of capital (increase in the return of investment opp. $q(x)$) ?

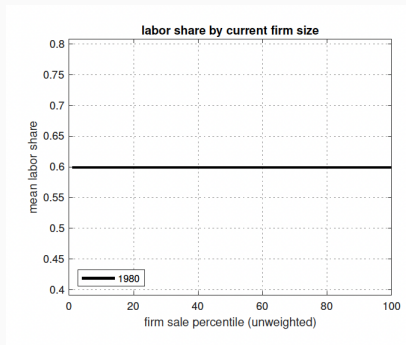
- Standard firm dynamics parametrization to match entry, exit, firm size, markup
- Set technology parameters to match key facts on manufacturing labor share:
 - task-substitution elasticity $\eta = 0.4$: match short-run K-L elasticity (Oberfield-Raval 2020)
 - specify $\psi_l(x) = \left(x^{\frac{1-\eta-\gamma}{\gamma}} - 1 \right)^{\frac{1}{1-\eta-\gamma}}$ and $\psi_k(x) = 1$
 $\eta + \gamma$ gives long-run aggregate K – L elasticity: target 1.35 (Hubmer, 2020; Karabarbounis & Neiman, 2014)
 - $d \ln q = 1.39$ to match observed decline in aggregate mf. labor share 1982–2012 (–17.8pp)
 - choose automation cost c_a to target change in median labor share (+3pp)
- What happens if we impose a uniform decrease in the rental price of capital (increase in the return of investment opp. $q(x)$) ?

Aggregate Labor Share Dynamics

- Economy starts in steady state in 1982
→ all firms have same labor share
- In terms of **aggregate factor shares**, fast transition
- In terms of **average firm**, slow transition

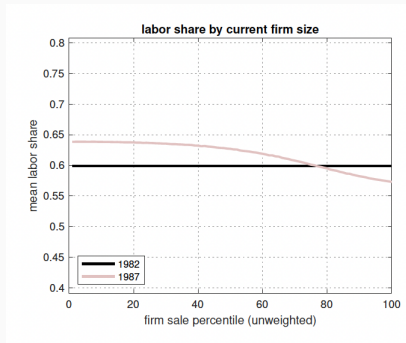


Firm-level Labor Share Dynamics



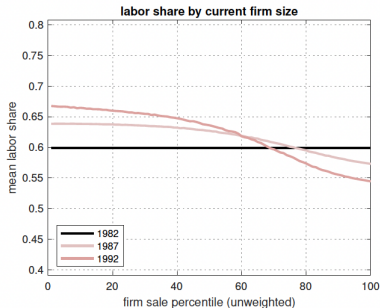
- In 1982 steady state, **uniform technology** and therefore **uniform labor share**

Firm-level Labor Share Dynamics



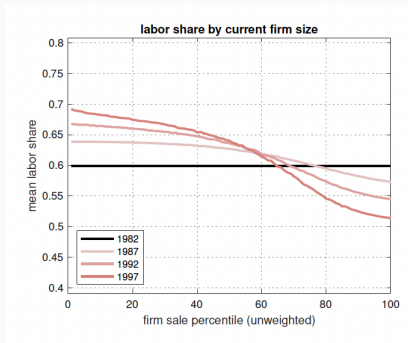
- In 1982 steady state, **uniform technology** and therefore **uniform labor share**
- As capital becomes cheaper and wages rise, **labor shares of small non-automating firms increase**

Firm-level Labor Share Dynamics



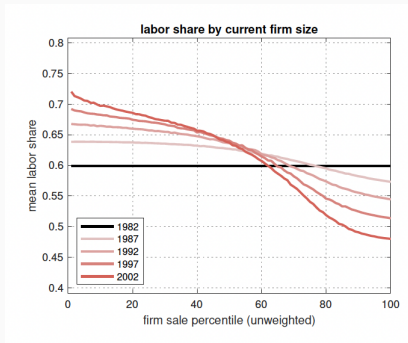
- In 1982 steady state, **uniform technology** and therefore **uniform labor share**
- As capital becomes cheaper and wages rise, **labor shares of small non-automating firms increase**
- **Firms that become large** reaching top sales percentiles are the ones **reducing their labor shares**

Firm-level Labor Share Dynamics



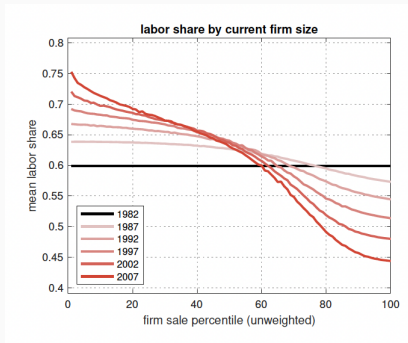
- In 1982 steady state, **uniform technology** and therefore **uniform labor share**
- As capital becomes cheaper and wages rise, **labor shares of small non-automating firms increase**
- **Firms that become large** reaching top sales percentiles are the ones **reducing their labor shares**

Firm-level Labor Share Dynamics



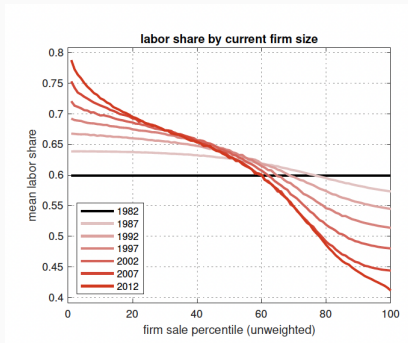
- In 1982 steady state, **uniform technology** and therefore **uniform labor share**
- As capital becomes cheaper and wages rise, **labor shares of small non-automating firms increase**
- **Firms that become large** reaching top sales percentiles are the ones **reducing their labor shares**

Firm-level Labor Share Dynamics



- In 1982 steady state, **uniform technology** and therefore **uniform labor share**
- As capital becomes cheaper and wages rise, **labor shares of small non-automating firms increase**
- **Firms that become large** reaching top sales percentiles are the ones **reducing their labor shares**

Firm-level Labor Share Dynamics



- In 1982 steady state, **uniform technology** and therefore **uniform labor share**
- As capital becomes cheaper and wages rise, **labor shares of small non-automating firms increase**
- **Firms that become large** reaching top sales percentiles are the ones **reducing their labor shares**

Decomposition of Labor Share in Manufacturing 1982–2012

	Data	CES Model
Δ Unweighted survivors' mean	-0.2	3.0
Δ Covariance	-18.7	-20.9
Entry	5.9	0.6
Exit	-5.5	-0.3
Δ Aggregate	-18.5	-17.6

- The labor share in an industry is given by $s_l = \sum_i w_i s_{il}$ where $s_{il} \rightarrow$ firm i LS and $w_i \rightarrow$ firm i share in VA
- Melitz-Polanec Decomposition of changes in s_i

$$\Delta s_l = \Delta \bar{s}_l + \Delta \text{cov}(w_i, s_{il}) + \text{entry} + \text{exit}$$

Unpacking the Changing Covariance Between Size and Labor Share

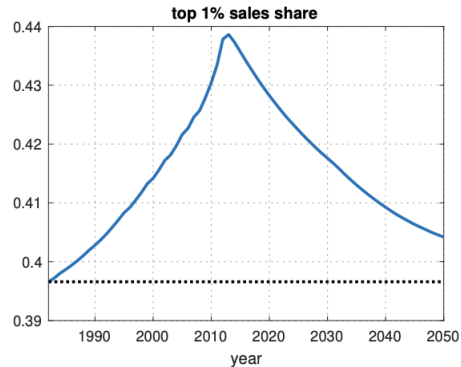
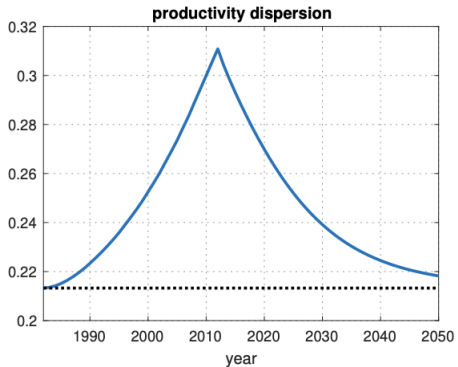
	Data (in p.p.)	CES Model (in p.p.)
Market share dynamics	4.7	0.0
Labor share dynamics*	-4.3	-4.1
Cross dynamics	-23.1	-13.7

- Covariance can be further decomposed

$$\Delta \text{cov}(w_i, s_{il}) = \text{cov}(w_i, \Delta s_{il}) + \text{cov}(\Delta w_i, s_{il}) + \text{cov}(\Delta w_i, \Delta s_{il})$$

- Uncertainty about exact contribution of these terms in data...

Productivity Dispersion and Sales Concentration



Summarizing Findings from CES Model

Simple firm dynamics model with endogenous automation:

1. matches aggregate and firm labor share dynamics
2. generates increasing concentration of sales and dispersion of productivity

Now: incorporate differences in markups to separate role of reallocation to large firms due to tighter competition from K—L substitution

**Model with non-CES demand to
compare effects of competition and
capital–labor substitution**

Size-dependent Markups and Rising Competition

- Non-CES demand system: Kimball aggregator $H(\cdot)$ implicitly defines aggregate output Y as the solution to

$$\int_{\theta} \lambda H\left(\frac{y(\theta)}{\lambda Y}\right) m(\theta) d\theta = 1$$

with λ as a(n exogenous) proxy for “market size”

- Normalizing price of final good to 1 yields demand function

$$y(\theta) = Y \lambda D\left(\frac{p(\theta)}{\rho}\right), \rho = \text{comp. price index} \neq 1, (H' = D^{-1})$$

- Key Assumptions: Marshall's Second Laws

$$-\frac{D'(x)}{D(x)} x \text{ greater than 1 and increasing in } x \quad (\text{markups higher and prices lower for prod. firms})$$

$$x + \frac{D(x)}{D'(x)} \text{ positive and log - concave} \quad ((\text{passthroughs lower for prod. firms}))$$

3 reasons: empirical backing, tractable way to generating pricing dynamics as in olig. comp. models, ensures that prod. firms are larger with higher markups and that more competition reallocates economic activity towards these firms

Size-dependent Markups and Rising Competition

- Non-CES demand system: Kimball aggregator $H(\cdot)$ implicitly defines aggregate output Y as the solution to

$$\int_{\theta} \lambda H\left(\frac{y(\theta)}{\lambda Y}\right) m(\theta) d\theta = 1$$

with λ as a(n exogenous) proxy for “market size”

- Normalizing price of final good to 1 yields demand function

$$y(\theta) = Y \lambda D\left(\frac{p(\theta)}{\rho}\right), \rho = \text{comp. price index} \neq 1, (H' = D^{-1})$$

- Key Assumptions: Marshall's Second Laws

$$-\frac{D'(x)}{D(x)} x \text{ greater than 1 and increasing in } x \quad (\text{markups higher and prices lower for prod. firms})$$

$$x + \frac{D(x)}{D'(x)} \text{ positive and log - concave} \quad ((\text{passthroughs lower for prod. firms}))$$

3 reasons: empirical backing, tractable way to generating pricing dynamics as in olig. comp. models, ensures that prod. firms are larger with higher markups and that more competition reallocates economic activity towards these firms

Size-dependent Markups and Rising Competition

- Non-CES demand system: Kimball aggregator $H(\cdot)$ implicitly defines aggregate output Y as the solution to

$$\int_{\theta} \lambda H\left(\frac{y(\theta)}{\lambda Y}\right) m(\theta) d\theta = 1$$

with λ as a(n exogenous) proxy for “market size”

- Normalizing price of final good to 1 yields demand function

$$y(\theta) = Y\lambda D\left(\frac{p(\theta)}{\rho}\right), \rho = \text{comp. price index} \neq 1, (H' = D^{-1})$$

- Key Assumptions: Marshall's Second Laws

$$-\frac{D'(x)}{D(x)} x \text{ greater than 1 and increasing in } x \quad (\text{markups higher and prices lower for prod. firms})$$

$$x + \frac{D(x)}{D'(x)} \text{ positive and log - concave} \quad ((\text{passthroughs lower for prod. firms}))$$

3 reasons: empirical backing, tractable way to generating pricing dynamics as in olig. comp. models, ensures that prod. firms are larger with higher markups and that more competition reallocates economic activity towards these firms

Size-dependent Markups and Rising Competition

- Non-CES demand system: Kimball aggregator $H(\cdot)$ implicitly defines aggregate output Y as the solution to

$$\int_{\theta} \lambda H\left(\frac{y(\theta)}{\lambda Y}\right) m(\theta) d\theta = 1$$

with λ as a(n exogenous) proxy for “market size”

- Normalizing price of final good to 1 yields demand function

$$y(\theta) = Y\lambda D\left(\frac{p(\theta)}{\rho}\right), \rho = \text{comp. price index} \neq 1, (H' = D^{-1})$$

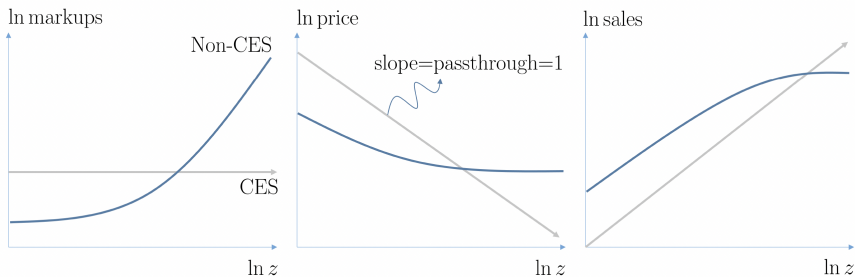
- Key Assumptions: Marshall's Second Laws

$$-\frac{D'(x)}{D(x)} x \text{ greater than 1 and increasing in } x \quad (\text{markups higher and prices lower for prod. firms})$$

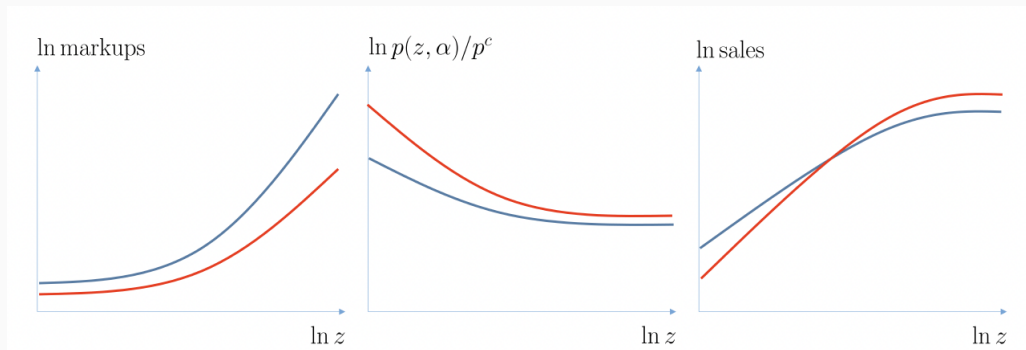
$$x + \frac{D(x)}{D'(x)} \text{ positive and log - concave} \quad ((\text{passthroughs lower for prod. firms}))$$

3 reasons: empirical backing, tractable way to generating pricing dynamics as in olig. comp. models, ensures that prod. firms are larger with higher markups and that more competition reallocates economic activity towards these firms

Steady State Markups, Pricing and Sales



Steady State Effects of Rising Competition, $\lambda \uparrow$



- $\lambda \uparrow$ increases market size, pushing firms towards elastic section of their demand curve
- reallocates economic activity towards top firms; but also lowers markups for all firms
- net increase in the aggregate markup if z has a log-convex distribution

- $H(\cdot)$: Klenow-Willis aggregator $\implies$ demand elasticity $:= \sigma \cdot \text{rel. quantity}^{-\nu/\sigma}$
 - $\sigma = 6.1$ (Manuf.) vs. $\sigma = 9.0$ (Retail)
 - $\nu/\sigma = 0.22$ (Manuf.) vs. $\nu/\sigma = 0.20$ (Retail)
- Productivity process given by

$$z = \exp\left(F_{Weibull(\zeta, n)}^{-1}(\Phi(\tilde{z}))\right), \tilde{z}' = \rho_z \tilde{z} + \varepsilon_z'$$

which ensures that

$$\ln(z) \sim \text{Weibull}(\zeta, n)$$

- $n = 0.78$ and $\zeta = 0.086$ (Manuf.) vs $n = 0.54$ and $\zeta = 0.023$ (Retail)
- Thus, prod. distribution more log-convex in retail than in manuf.

Transitional Dynamics: Manufacturing 1982–2012

Automation and rise in
comp. calibrated to match

- decline in labor share
- observed rise in concentration

Inferred shocks

- $d \ln q = 1.40$
- $d \ln \lambda = 0.04$

		MODEL			
		DATA	FULL	q -SHOCK	λ -SHOCK
	Δ aggregate LS ^a	−17.8	−17.3	−16.3	−0.2
	Δ median LS ^a	3.0	2.6	1.4	0.2
	Δ CR4 ^b	6.0	6.0	3.8	1.9
	$\Delta \ln$ agg. markup	.	1.1	1.0	0.1

In percentage points. [a] Kehrig–Vincent (2020). [b] Autor et al (2020): Average manuf. industry sales concentration.

Transitional Dynamics: Retail 1982–2012

Automation and rise in
comp. calibrated to match

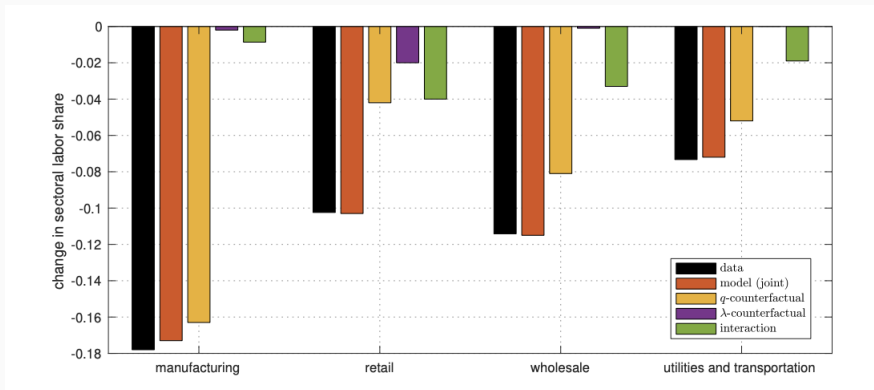
- decline in labor share
- observed rise in concentration

Inferred shocks

- $d \ln q = 0.75$ (1.40 in manuf)
- $d \ln \lambda = 0.41$ (0.04 in manuf)

		DATA	MODEL		
			FULL	q -SHOCK	λ -SHOCK
	Δ aggregate LS ^a	−10.2	−10.3	−4.2	−2.0
	Δ uw. mean LS ^b	4.4	4.2	−2.3	1.4
	Δ CR4 ^b	14.0	11.5	0.1	10.8
	Δ CR20 ^b	16.3	20.4	0.3	18.8
	$\Delta \ln$ agg. markup	.	3.3	0.1	2.9
In percentage points. [a] BEA Multifactor Productivity Tables. [b] Autor et al (2020).					

Model-based decomposition of sectoral labor share changes



Model-free bounds on effect of reallocation and changes in markups on labor share

Model-free Bounds on Markup Changes and Reallocation

- So far, only accounted for **size-dependent component** of markups
 - In the data, much more markup variation
- Now allow for more general markup variation and reallocation (Baqaei–Farhi, 2020)
 - Greater generality comes at cost of requiring firm-level markup estimates.
 - Strong assumptions are inevitable (Compustat; prices)
- The authors follow the literature (DeLoecker–Eeckhout–Unger, 2020) in using the **production function** approach to recover output elasticities ε_{vft} wrt expenditures in variable inputs v
 - $\implies$ markup $\mu_{ft} = \varepsilon_{vft}/s_{vftt}$ for firm t in industry i
 - Contribution: the authors allow technology ε_{vft} to vary by time period, industry, and firm size

Model-free Bounds on Markup Changes and Reallocation

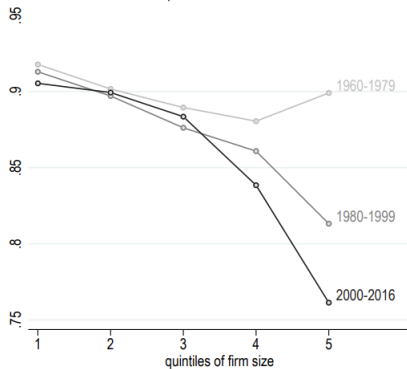
- So far, only accounted for **size-dependent component** of markups
 - In the data, much more markup variation
- Now allow for more general markup variation and reallocation (Baqaei–Farhi, 2020)
 - Greater generality comes at cost of requiring firm-level markup estimates.
 - Strong assumptions are inevitable (Compustat; prices)
- The authors follow the literature (DeLoecker–Eeckhout–Unger, 2020) in using the **production function** approach to recover output elasticities ε_{vft} wrt expenditures in variable inputs v
 $\implies$ markup $\mu_{ft} = \varepsilon_{vft}/s_{vftt}$ for firm t in industry i
 - Contribution: the authors allow technology ε_{vft} to vary by time period, industry, and firm size

Model-free Bounds on Markup Changes and Reallocation

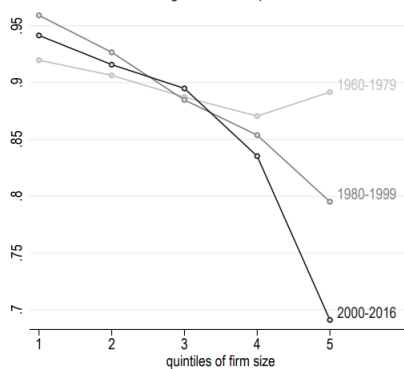
- So far, only accounted for **size-dependent component** of markups
 - In the data, much more markup variation
- Now allow for more general markup variation and reallocation (Baqaei–Farhi, 2020)
 - Greater generality comes at cost of requiring firm-level markup estimates.
 - Strong assumptions are inevitable (Compustat; prices)
- The authors follow the literature (DeLoecker–Eeckhout–Unger, 2020) in using the **production function** approach to recover output elasticities ε_{vft} wrt expenditures in variable inputs v
 - ⇒ markup $\mu_{ft} = \varepsilon_{vft}/s_{vftt}$ for firm t in industry i
 - Contribution: the authors allow technology ε_{vft} to vary by time period, industry, and firm size

Clockwise Rotation in Elasticities

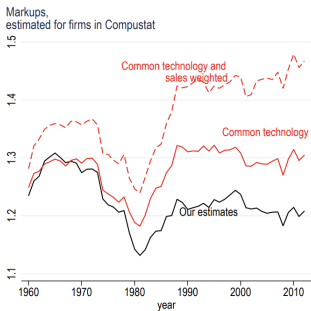
Output elasticity wrt variable inputs,
estimated for firms in Compustat



Output elasticity wrt variable inputs,
estimated for manufacturing firms in Compustat



Aggregate Markup – $\mu_t = 1 / \sum_f w_{ft} \cdot (1/\mu_{ft})$



- Common technology, sales weighting (DLEU 2020): replicate strong increase in aggregate markup
- Common technology, cost-weighting (Edmond, Midrigan, Xu, 2021; BF): mild increase
- Author's estimate with heterogeneous technology and cost-weighting: no trend

Contribution to Labor Share Decline



- In this figure, the authors plot the cumulative percentage change in $1/\mu_t$ between 1960 and each year aggregated across all industries

Conclusion and Comments

Concluding Remarks

- Model of K–L substitution with a fixed cost per task matches firm-level facts on the decline of labor share
- When accounting for rising competition and reallocation to more productive firms, K–L substitution retains an important role
 - explains 90% of the decline in labor share in manufacturing
 - explains 50% of the decline in labor share in retail
- Model-free bounds on role of competition and concentration support these findings
 - highlights importance of allowing for differences in technology across firms

- It is a nice paper, the idea is clear and interesting while the execution is also well done
- As any paper, there are some concerns, they don't invalidate the analysis but they warrant a deeper look into what is going into the quantitative results
- Doubts about calibration:
 - Validate q and λ **outside** of the model
 - Test the model against untargeted moments
 - Check sensitivity of moments with regards to changes in the parameters chosen
- Connect empirical test with model
 - Model has enough structure to run the production function estimation **within** its structure
 - What does it tell us about the parameters that are being chosen?